

Netis Solar Intelligence Platform Roadmap

Phase 0 – Business Discovery (2 Weeks)

Objectives

- Define target customers: industrial, commercial, office buildings, warehouses, malls, and similar segments.
- Identify countries and regions to cover first.
- Assess available imagery sources.
- Define KPIs for success.

Deliverables: Product Vision; Business Requirements; Functional Specifications; Technical Architecture; ROI Model.

Phase 1 – GIS Foundation (3 Weeks)

Objectives: Build the geospatial data platform.

Data Sources: Satellite imagery; OpenStreetMap; government cadastral data; building footprints; road network; administrative boundaries.

Database: PostGIS, including tables for Buildings, Businesses, Imagery, Roofs, Solar Potential, Customers, and Sales Opportunities.

Deliverables: ✓ Interactive Map; ✓ Building Database; ✓ GIS APIs.

Phase 2 – AI Rooftop Detection (4 Weeks)

Objectives: Automatically detect rooftops.

AI Models: YOLOv11; Segment Anything (SAM).

Outputs: Roof Polygon; Roof Area; Roof Shape; Building ID.

Example: Building roof area: 785 m².

Deliverables: ✓ Automatic rooftop segmentation; ✓ Roof polygons stored in PostGIS.

Phase 3 – Roof Intelligence (4 Weeks)

The AI becomes more intelligent by detecting rooftop elements and calculating usable surface area.

Detect: HVAC units; skylights; water tanks; existing solar panels; chimneys.

Compute: Total roof; obstacle area; available area.

Example: Total roof area: 830 m²; available area: 675 m².

Deliverables: ✓ Usable rooftop area; ✓ Obstacle detection.

Phase 4 – Solar Potential Engine (3 Weeks)

Using available roof area, estimate the number of panels, installed power in kWp, annual production, annual savings, CO₂ reduction, and ROI.

Libraries and data: pvlib; PVGIS data; local irradiation data.

Output example: Capacity: 180 kWp; production: 300 MWh/year; savings: \$42,000/year; CO₂ reduction: 120 tons/year.

Deliverables: ✓ Solar simulation engine.

Phase 5 – Business Intelligence Enrichment (4 Weeks)

This phase transforms buildings into sales leads by enriching each location with business data.

Retrieve: Company name; industry; number of employees; phone; email; website; revenue, if available; decision makers; business activity.

Potential sources: Government business registry; OpenStreetMap; Google Places; commercial business databases; internal CRM.

Data relationship: Business → Building → Roof → Solar Potential.

Deliverables: ✓ Business lookup engine.

Phase 6 – AI Lead Scoring (2 Weeks)

Every building receives a score based on solar potential and sales attractiveness.

Factor	Weight
Roof Size	25%
Business Size	20%
Estimated Consumption	20%
Industry	15%
Roof Orientation	10%
Distance	10%

Output example: Lead Score: 94/100; Priority: Excellent, Good, Average, or Low.

Deliverables: ✓ Lead ranking.

Phase 7 – Sales CRM (3 Weeks)

Interactive dashboard example: Click a building to view ABC Manufacturing, roof area of 720 m², available area of 610 m², solar potential of 160 kWp, estimated savings of \$38,000/year, phone, email, and current status such as Not Contacted.

CRM features: Assign salesperson; visit planning; notes; pipeline; follow-up; opportunity status.

Deliverables: ✓ Sales Portal.

Phase 8 – AI Assistant (3 Weeks)

The LLM understands every opportunity and helps sales and technical users act faster.

Example prompts: Summarize this prospect; estimate proposal value; explain why this building scores only 60; draft the first sales email; generate a technical report.

The assistant reads: Roof geometry; solar potential; business information; CRM history.

Deliverables: ✓ AI Sales Copilot.

Phase 9 – Proposal Generator (3 Weeks)

Generate professional PDF proposals automatically in one click.

Contents: Company profile; satellite view; roof analysis; solar layout; energy production; financial analysis; ROI; CO₂ savings; pricing; financing options.

Deliverables: ✓ Proposal generator.

Phase 10 – Executive Dashboard (2 Weeks)

Power BI dashboards provide executive visibility across sales, operations, financial performance, and AI performance.

- **Sales KPIs:** Opportunities; proposal value; conversion rate; sales pipeline; revenue forecast.
- **Operations KPIs:** Roofs analyzed; area analyzed; solar capacity identified; potential MW.
- **Financial KPIs:** Expected revenue; ROI; average installation value; monthly sales.
- **AI KPIs:** Detection accuracy; AI confidence; proposal generation time.

Deliverables: ✓ CEO Dashboard.

Phase 11 – Advanced AI (Future)

Future features: Shadow detection for trees, buildings, poles, and mountains; roof tilt estimation for slope, orientation, and azimuth; automatic panel layout; drone inspection; thermal camera analysis; predictive maintenance; and portfolio optimization.

- **Automatic Panel Layout:** AI places solar panels automatically.
- **Drone Inspection:** Upload drone imagery for automatic inspection and defect detection.
- **Thermal Camera Analysis:** Detect hot spots, broken panels, and dirt accumulation.
- **Predictive Maintenance:** Forecast cleaning, inverter failures, and panel degradation.
- **Portfolio Optimizer:** For companies owning multiple buildings, AI recommends which roofs to equip first, expected payback, and optimal installation sequence.

High-Level Architecture

Satellite Imagery → AI Rooftop Segmentation → Roof Intelligence Engine → Solar Potential Engine and Business Enrichment → Lead Scoring Engine → GIS Portal, CRM, and AI Assistant → Proposal Generator → Power BI Executive Dashboards.

Success Metrics

KPI	Target
Rooftop detection precision	>95%
Rooftop area estimation error	<5%
Business matching rate	>90%
Proposal generation time	<2 minutes
Lead qualification time	Reduced by 80%
Sales conversion improvement	+20–30% over current process
Time from prospect identification to first contact	<24 hours

This roadmap provides a practical path to an MVP in approximately 4 months through Phases 0–4, followed by business enrichment, CRM, AI assistance, proposal automation, and executive analytics that can evolve into a comprehensive Solar Intelligence Platform.

